

# The Morse Hamiltonian: Exact Classical Action and Complete Quantum Bound Spectrum

Route-A source-local certificate HCS-C334

3 September 2026

## Abstract

We solve the classical and quantum one-dimensional Morse Hamiltonian under one convention lock. Every bounded classical orbit has an elementary action and period, while canonical quantization has a finite, explicitly exhausted generalized-Laguerre point spectrum.

## 1 Frozen Hamiltonian and result

Let

$$V(x) = D(e^{-2ax} - 2e^{-ax}), \quad H_{\text{cl}}(x, p) = \frac{p^2}{2m} + V(x), \quad (1)$$

where  $m, D, a > 0$ . The potential has minimum  $-D$  at zero, tends to zero from below as  $x \rightarrow +\infty$ , and tends to  $+\infty$  as  $x \rightarrow -\infty$ . For a closed regular energy curve we use the normalized action

$$J(E) = \frac{1}{2\pi} \oint p \, dx. \quad (2)$$

For  $\hbar > 0$ , let  $H_q$  be the self-adjoint operator associated with the lower-bounded closed form of

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \quad \text{on } L^2(\mathbb{R}), \quad \lambda = \frac{\sqrt{2mD}}{a\hbar}.$$

**Theorem 1** (Classical action and quantum closure). *For every  $-D < E < 0$ , the classical energy curve is periodic and*

$$J(E) = \frac{\sqrt{2mD}}{a} (1 - \sqrt{-E/D}), \quad T(E) = \frac{2\pi}{a\sqrt{-2E/m}}, \quad J'(E) = \frac{T(E)}{2\pi}. \quad (3)$$

*The essential spectrum of  $H_q$  is  $[0, \infty)$ . Its point spectrum is exactly the finite set*

$$E_n = -\frac{a^2 \hbar^2}{2m} (\lambda - n - \tfrac{1}{2})^2, \quad n \in \mathbb{N}_0, \quad n < \lambda - \tfrac{1}{2}, \quad (4)$$

*and every eigenvalue is simple. With  $z = 2\lambda e^{-ax}$ , a corresponding eigenfunction is, up to normalization,*

$$\psi_n(x) = z^{\lambda-n-1/2} e^{-z/2} L_n^{(2\lambda-2n-1)}(z). \quad (5)$$

*There are  $\max(0, \lceil \lambda - 1/2 \rceil)$  bound states. Equality in the cutoff produces a zero-energy constant tail, not an  $L^2$  eigenfunction; there are no nonnegative eigenvalues.*

The phrase *Morse classical action owner* is the round-zero certificate: it fixes the model, the  $2\pi$  normalization, and the strict spectral statement before any finite computation.

## References

- [1] P. M. Morse, *Diatomic Molecules According to the Wave Mechanics. II. Vibrational Levels*, Phys. Rev. 34 (1929), 57, [doi:10.1103/PhysRev.34.57](https://doi.org/10.1103/PhysRev.34.57).
- [2] NIST Digital Library of Mathematical Functions, *Laguerre polynomials*, <https://dlmf.nist.gov/18.5>.
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